

Red Sand

Propellant Lead • Arcadia Rise

15 SOLS • 61 STOPS • GRADE 12 • REDSAND

THE OPENING CARD

Everyone here goes home on a rocket, and the rocket can only leave when Mars and Earth line up. That happens once every twenty-six months. The next one is sixteen months off. The rocket is already standing on the pad, four hundred metres from your door. It has to lift off full, and nothing is being flown out from Earth to fill it. The 6.6 tonnes of methane and 23 of oxygen have to be made here, out of the Martian air and the ground under it. So far the plant has made 3.9. You are the propellant lead at Arcadia Rise, so every kilogram loaded into that rocket is signed for by you. The plant has run under its rated output every sol since spring. This morning Ingrid Sundqvist wants the reactor forty degrees hotter, and Tomás Herrera says a hotter reactor hands back less methane on every pass. Six people fly home on whatever those two can be got to agree to make.

2700 kilograms short

BEFORE THE SOL — GREET

Six people, and the next crew is eighteen months away

There are six of you and no relief. Sundqvist, the plant production lead, wants you known to as many of them as you can before the first sol of production, because a disagreement here is not settled by anybody leaving the room — there is nowhere to go, and the transfer window is 486 sols out.

PLAN CARD 126W

Sol 291, and the reconciliation of the ascent vehicle's tanks was read out at breakfast this morning. 3.9 tonnes of methane are aboard and 6.6 have to be aboard by the sol the transfer window opens, 195 sols from now, and the plant's present output does not reach it. Ingrid Sundqvist, the plant production lead, wants the reactor set point raised 40 degrees tonight. Tomás Herrera, who owns the reactor loop and the specification the propellant has to meet, says a hotter bed gives back less methane on every pass. Today you walk the loop end to end and decide what is holding the output down. Whatever the plant misses this fortnight is carried by the sols that are left, and there are 195 of those.

BRIEFING 15W

The tank reconciliation was read at breakfast and the plant does not reach the number.

WHAT YOU DECIDE 14W

Establish what the plant is actually limited by before anybody changes a set point.

1.1 What the loop settles at

REACTOR HALL · A MODULE · PROTOCOL

SCENE 35W

The loading projection is still 2,700 kilograms short. Herrera has stopped accepting slogans like 'run it hotter.' Four proposed changes are on the reactor board. He wants each effect agreed before anyone touches the loop.

GUIDE 62W

Four changes to the loop, and four effects. Pair them by asking what each change actually touches. The number of gas molecules on each side? The heat, which is one of the products here? A partial pressure of something taking part? Or nothing at all? One of these raises the total pressure and no partial pressure, so the mixture never notices it.

ASK 15W

Match each change to the loop with what it does to the methane at equilibrium.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a reaction at equilibrium is still running in both directions · pressure and temperature are the two settings on a gas-phase reactor

VERDICT 83W

Each of these is the same principle applied to a different quantity. Pressure counts gas molecules, and five go in where three come out, so squeezing the loop favours the smaller side. Temperature is different in kind: heat is one of the products here, so the equilibrium constant itself falls as the bed is heated. Condensing the water changes the ratio without touching the constant. Argon raises the total pressure and no partial pressure, so the mixture never notices that it is there.

TAKEAWAY 17W

Pressure, temperature, product removal and an inert gas do not disturb an equilibrium in the same way.

1.2 Rate is not yield

CATALYST BAY · A PERSON · CHOICE

SCENE 37W

Sundqvist wants the bed at 320 °C tonight, up from 280. Ravneet Kaur, the catalysis engineer, has the temperature profile and the production projection side by side. One setting may rescue the schedule or deepen the shortfall.

GUIDE 56W

Four options, and temperature moves two different quantities. Ask of each whether it keeps them separate. The rate constant rises because more collisions clear the barrier. The equilibrium constant of a heat-releasing reaction falls. So a hotter bed can reach a lower ceiling faster, and two of these options treat speed and yield as one thing.

ASK 17W

The bed is run 40 degrees hotter at the same pressure and the same flow. What happens?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

heating a reaction makes it go faster · an exothermic reaction gives out heat as it proceeds

VERDICT 73W

Temperature moves two different quantities in opposite directions. The rate constant rises because more collisions can clear the activation barrier. That is a kinetic effect. For this exothermic reaction, the equilibrium constant falls as temperature rises. That is a thermodynamic effect. A hotter bed can therefore reach a lower methane ceiling more quickly. Whether the trade helps depends on recycle, residence time and how much unreacted feed the loop can send around again.

TAKEAWAY 14W

Temperature can increase reaction rate while decreasing the equilibrium yield of an exothermic reaction.

1.3 What freezes before methane condenses

COLD END · A MODULE · CHOICE

SCENE 34W

The cold line takes the product stream from 10 degrees down to 115 kelvin in 1 run of pipe. Mei-Ling Cho, the cryogenics engineer, will not accept a stream that has skipped the drier.

GUIDE 63W

Four reasons to clean the stream first. Ask of each whether it names a temperature at which something leaves the gas. Water goes near 273 K and carbon dioxide near 195 K, both of which the line passes through on its way down. Solid on an exchanger surface narrows the passage, and from outside that looks like a valve or a refrigeration fault.

ASK 18W

Why does the stream pass a drier and a carbon dioxide trap before it reaches the cold line?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

a substance leaves the gas phase at a temperature that depends on the pressure · liquid methane needs about 112 K at loop pressure · intermolecular forces, and why they set a boiling point — taken as read

VERDICT 80W

Every substance in the stream leaves the gas phase at its own temperature. Water freezes near 273 K. Carbon dioxide can deposit directly as a solid near 195 K at these pressures. Methane remains gaseous until the stream is much colder, near its condensation temperature. If water and carbon dioxide are not removed first, both

can become solids on exchanger surfaces. A growing solid layer narrows the passage and can mimic a valve or refrigeration fault from outside the pipe.

TAKEAWAY 21W

A cold line is a series of temperatures, and every substance in the stream leaves the gas at its own one.

END OF THE SOL 15W

A plant has a speed and a ceiling, and they are not the same limit.

Run it hotter

BEFORE THE SOL — TRIAL

Walk the plant while your suit is fresh

Nine modules buried to the eaves along one track, and an ascent vehicle four hundred metres past the last of them. Walk the whole line once early, on a clear sol, because the next time you do it will be in a dust storm on a suit that has been out for three hours already.

PLAN CARD 125W

Sol 292, and the bed went to 320 degrees overnight on Sundqvist's authority. It made 11.4 kilograms of methane against 9.7 the sol before, and the recycle loop took back most of what the hotter pass gave away. Sundqvist is right this sol, and what she is right about is narrow: the bed had been running far below the temperature it was designed for, and a loop that recycles can afford a poor pass. Today you go to the electrolysis hall and settle what a stack actually converts, since hydrogen is the other half of every kilogram of methane. The heating hour the plant borrowed to do it came off Yusuf Demir's habitat margin, and the life support officer has said so in the log.

BRIEFING 16W

The hot run made its number overnight, and the argument has moved to the electrolysis hall.

WHAT YOU DECIDE 19W

Find what else the same power could be doing, now that the bed is where it should have been.

2.1 How long a molecule stays in the bed

CATALYST BAY • A MODULE • BALLPARK

SCENE 35W

Kaur wants the contact time worked out at the new flow before anything else is changed. The bed holds 340 litres of packed catalyst and the loop is passing 85 litres a minute through it.

GUIDE 60W

Five numbers, and three of them describe the conditions rather than the geometry: the pressure, the temperature, and the feed ratio. Ask of each whether this time depends on it. Residence time is how long the gas stays with the catalyst. Push more gas through the same bed and that contact shortens, so flow and conversion move against each other.

ASK 11W

Estimate how long a molecule of gas spends inside the bed.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a flow quoted in litres a minute is a volume passing each minute • gas volumes depend on the pressure and temperature they are quoted at • rate constants, and how steeply they climb with temperature — taken as read • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 75W

Residence time is the contact the reaction gets. A rate constant says how quickly reaction occurs at the stated conditions. Residence time says how long the gas remains with the catalyst. Conversion depends on both. Raising throughput by pushing more gas through the same bed shortens that contact. The plant can therefore increase flow while lowering conversion in one move. Rate and residence time must be judged together before a higher production setting is accepted.

TAKEAWAY 18W

Conversion depends on how fast the reaction runs and on how long it is allowed to run for.

2.2 Which electrode does which job

ELECTROLYSIS HALL • A PERSON • PROTOCOL

SCENE 33W

The hall runs two kinds of cell: 1 splits water, the other splits carbon dioxide at 800 degrees. Rosalind Achebe, the power and electrolysis engineer, has the four parts listed on the board.

GUIDE 58W

Four parts of the cell, and four jobs. Pair them by asking which way the electrons go. In means reduction; out means oxidation. Then ask what the electrolyte is for: it carries ions and refuses electrons, which is what forces the circuit through the wire. And one item on the list decides an amount rather than a direction.

ASK 10W

Match each part of the cell to what it does.

BEHIND THE BUTTON 192W

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TAKES AS READ

oxidation is loss of electrons and reduction is gain of electrons • a current is a flow of charge • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 78W

Electrolysis separates the two halves of one electron-transfer reaction. Electrons enter at the cathode, where reduction occurs. Electrons leave at the anode, where oxidation occurs. The solid electrolyte carries ions between those regions without carrying electrons through the same path. That separation keeps oxygen away from the fuel-side product. Current measures how quickly charge is delivered through the external circuit. Faraday's law then connects the total charge passed to the maximum amount of product the cell can make.

TAKEAWAY 20W

The two halves of one electron transfer can be run in separate places if something carries an ion between them.

2.3 What has to be paid for

PLANT CONTROL · A MODULE · BALLPARK

SCENE 34W

Demir has the sol's energy ledger open on the desk. The reactor releases heat while it runs, while electrolysis takes most of the array output. He wants the difference stated plainly in the log.

GUIDE 69W

Put a number on the direction. Water splitting has a known enthalpy change and a known entropy change, and the free-energy change at the cell's working temperature is what says whether it runs on its own. Watch the units: the entropy term is in joules per kelvin per mole and the enthalpy in kilojoules per mole, so one of the tiles is there to be converted rather than added.

ASK 16W

What is the free-energy change for splitting a mole of water at the cell's working temperature?

BEHIND THE BUTTON 165W

What the two terms are doing. The enthalpy term is the heat the change takes in or gives out. The entropy term is how much more, or less, spread out the matter and its energy end up, and it is multiplied by the temperature — which is why heating something can turn a change that does not run into one that does.

Why a positive answer means a bill. A positive free-energy change means the products sit above the reactants, so the energy has to come from somewhere outside the reaction. In this plant it comes from the array, through the cell, at a price in kilowatt-hours that the ledger on Demir's desk has to carry.

Why the reverse is free. Methane assembly is water splitting run backwards, and it gives out what the splitting takes in. That is one number with two signs, not two independent facts, and it is why the plant can run one step on its own and never the other.

TAKES AS READ

some reactions give out heat and others take it in · electrical energy can be counted in kilowatt-hours

VERDICT 157W

Direction is set by the free-energy change: $\Delta G = \Delta H - T\Delta S$. Splitting a mole of liquid water takes in 286 kJ, and the products are two gases from a liquid, so the entropy change is positive at +163 J per kelvin per mole. At the cell's 298 K the entropy term is $298 \times 0.163 = 48.6$ kJ per mole, which leaves $\Delta G = 286 - 48.6 \approx +237$ kJ per mole. Positive means the products sit above the reactants and nothing about the equipment changes that — the energy has to arrive from outside, which is what the array is for. Methane assembly is the same number with the opposite sign, which is why one step runs on its own and the other never will. And note what the

entropy term does not do here: it is nowhere near large enough to turn this one spontaneous, even at the highest temperature the cell could survive.

TAKEAWAY 23W

Whether a change runs on its own and how fast it goes are two separate questions, and one of them has a bill.

2.4 Worth digging, or worth leaving

WATER PLANT · A MODULE · BELT

SCENE 42W

The hopper takes one charge at a time and the digging arm has brought in more than it can process. Some of what is out there gives up water when it is heated. The rest costs the same power and returns nothing.

GUIDE 45W

Two bins. What matters is whether the material holds water that heat can drive off — hydrated minerals and buried ice do, and dry basaltic dust does not, however promising it looks. Sort on what the sample gives up, not on where it came from.

ASK 13W

Send each sample to the bin that says whether heating it returns water.

BEHIND THE BUTTON 87W

Why this is the plant's first decision. Every charge costs the same heating energy whatever is in it, and energy is the thing this station has least of. A hopper charged with dry dust has spent a sol's margin on nothing.

What holds water on Mars. Hydrated sulphates and clays hold it chemically, and it comes off at a few hundred degrees. Buried ice holds it as ice. Weathered basalt, glass and windblown fines hold almost none, and they are the majority of what the arm reaches.

TAKES AS READ

some minerals hold water that heating can drive off · intermolecular forces, and why they set a boiling point — taken as read

VERDICT 154W

Water is the input the whole plant runs on, and heating is how it is recovered, so the only question worth asking of a sample is whether heat gets water out of it.

Hydrated sulphates and clays do: the water is held in the mineral structure and comes off at a few hundred degrees. Buried ice does, obviously and at far lower cost. Weathered basalt, volcanic glass and the windblown fines that cover most of the site hold almost nothing, and they look exactly like the material that does — which is why the sorting happens at the yard rather than at the hopper. The cost of getting it wrong is not the sample. It is the charge: the heater draws the same power whatever is in it, energy is the scarcest thing on this station, and a hopper of dry dust has spent a sol's margin to produce steam that was never there.

TAKEAWAY 15W

Every charge costs the same energy, so what is in it is the whole decision.

END OF THE SOL 18W

A loop that recycles can afford a poor pass, and a plant with no hydrogen cannot afford anything.

The line that stopped passing gas

BEFORE THE SOL — FOLLOW

Cho walks the cold end once

The liquefier's cold end is the part of the plant nobody has a diagram of at the level that matters — which joint frosts first, which line ticks when it is cooling. Cho, the cryogenics engineer, walks it once. Stay with her and keep off the lines; a boot on a cold transfer line is a leak you cannot weld out here.

PLAN CARD 127W

Sol 293, and the cold end stopped delivering liquid 3 hours after the hot run began. The loop is still making methane and none of it is reaching a tank, so the sol's production is standing in a pipe at 12 bar. Cho has a flow that fell away over three hours rather than stopping at once, which she says rules out most of the simple explanations. Herrera is right this sol: the hotter bed put more water into the product stream, and the water went somewhere. Today you find where it went, work out why the hot pass gave back less methane, and follow the water back up the plant to the ground it came out of. Nothing else runs while the cold end is blocked.

BRIEFING 12W

The cold end stopped delivering liquid three hours into the hot run.

WHAT YOU DECIDE 16W

Find what closed the cold line, and what the hot run has to do with it.

3.1 Reading a blockage from outside the pipe

COLD END · A MODULE · DIAGNOSIS

SCENE 35W

Five readings are up on the cold end panel. Two of them are alarming, three are entirely ordinary, and Cho says the ordinary ones are the reason most of the obvious answers do not survive.

GUIDE 58W

Five readings, and the quiet ones do the work. A cold head at its set point has not lost capacity. A steady jacket pressure is intact insulation. Ask of each candidate how many of the five it fits. And note the shape of the flow trace: a shut valve stops flow in a second, while ice takes hours.

ASK 9W

Which explanation fits every reading, not the loudest one?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a refrigerator that has reached its set point is doing its job · water freezes at a far higher temperature than methane condenses at · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 84W

The quiet readings do the work. A cold head sitting at its set point has not lost capacity, and a steady jacket pressure is a jacket that has not failed. That leaves a blockage, and the shape of the flow trace says which kind: a shut valve stops the flow in a second, while ice builds up over hours and takes the flow down with it. The drier outlet at 900 ppm names the material. Upstream pressure rising against a growing plug completes it.

TAKEAWAY 11W

The readings that are normal are what rule an explanation out.

3.2 Why the hotter pass gave less

REACTOR HALL · A PERSON · BALLPARK

SCENE 32W

The hot run's assay is on Herrera's desk. Conversion per pass fell from 71% to 64% while throughput rose. He wants the reason written down before anyone argues for another temperature increase.

GUIDE 57W

The assay gives the outlet partial pressures, so the reaction quotient can be computed rather than argued about. Build Q from the four partial pressures, each raised to its own coefficient, and then compare it with the equilibrium constant at each of the two temperatures. Two of the tiles are the same pressures without their exponents applied.

ASK 19W

Compute the reaction quotient at the hot run's outlet, and say which way that mixture still has to go.

BEHIND THE BUTTON 158W

What Q is, and what it is not. Q has the same form as K and is evaluated wherever the mixture happens to be, so it is a description of the present rather than a constant. Comparing the two says which way the reaction still has to go, and nothing about how fast it will get there.

Why the exponents are not decoration. Every partial pressure enters raised to its coefficient in the balanced equation, and hydrogen's is four. Leaving it out changes Q by two orders of magnitude here, which is the difference between a mixture that is past its ceiling and one that is nowhere near it.

Why the hot run had two things wrong with it. Temperature moved the ceiling and throughput moved the residence time, and the assay alone cannot separate them. What it can settle is the ceiling: K at the hotter temperature is smaller, and the outlet mixture is already above it.

TAKES AS READ

the equilibrium constant is fixed at a given temperature · the Sabatier reaction gives out heat

VERDICT 130W

The Sabatier reaction is $\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$, so $Q = p(\text{CH}_4) \times p(\text{H}_2\text{O})^2 \div (p(\text{CO}_2) \times p(\text{H}_2)^4)$. At the hot outlet that is $0.42 \times 0.7056 \div (0.10 \times 0.0105) \approx 283$. K is 190 at 573 K and about 1,050 at 533 K, because the reaction gives out heat and heating an exothermic equilibrium lowers its constant. So the same outlet mixture is above the ceiling at the hot setting — Q exceeds K , and the reactor would run backwards to reach balance — and comfortably below it at the cooler setting. That settles the thermodynamic half of Herrera's question without settling the kinetic half: the higher throughput also shortened residence time, and only a flow test at fixed temperature and pressure separates those two.

TAKEAWAY 22W

Heating an exothermic reaction lowers its equilibrium product yield, while the observed reactor conversion can also depend on kinetics and residence time.

3.3 Getting water out of frozen ground

WATER PLANT · A MODULE · SEQUENCE

SCENE 33W

Femi Abiola, the regolith and water lead, has the plant's four steps written on cards. The morning crew disagreed about where the extra water appeared. He wants the cards restored to process order.

GUIDE 59W

All four steps will happen, so ask what each one needs from the step above it. Ice that melts in the bucket runs into the ground. Vapour released outside a sealed vessel goes into an atmosphere at six millibars. And an ion-exchange bed can only be fed liquid water. Out of order, each step has nothing to work on.

ASK 8W

Order the steps the water plant runs in.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

warming ice-bearing soil releases water as vapour at low pressure · dissolved salts stay behind when water evaporates · intermolecular forces, and why they set a boiling point — taken as read

VERDICT 85W

The order is set by what each step needs from the one above it. Ice that melts in the bucket runs into the ground and is lost, so the load stays frozen until it is inside something sealed. Vapour has to be captured where it is released or it goes into the atmosphere at six millibars. Condensing gives liquid water, and only liquid water can be run through an ion-exchange bed. Take the steps out of order and each one has nothing to work on.

TAKEAWAY 14W

Every step in the water plant fixes something the step after it cannot survive.

3.4 Rate is not yield — Review

CATALYST BAY · A MODULE · CALLBACK · CHOICE

SCENE 29W

The greenhouse reactor is running cold and slow. Its operator proposes 40 degrees more, and the same argument that ran through the propellant plant a fortnight ago starts again.

GUIDE 56W

Four pairs of statements, and each pair makes two claims. Ask of each whether the two halves are allowed to disagree. Temperature is carried by two quantities that move in opposite directions for a heat-releasing reaction. So the pair that has them both rising cannot be right, and neither can the pair that has neither moving.

ASK 15W

A reaction that gives out heat is run hotter. Which pair of statements is right?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

heating raises the fraction of collisions that clear the barrier · an exothermic reaction's balance is pushed back by heating

VERDICT 76W

Two quantities carry the temperature and they carry it in opposite directions. The rate constant rises steeply, since more collisions arrive with enough energy to clear the barrier. The equilibrium constant of a reaction that gives out heat falls, since heat behaves as one of its products. Every exothermic industrial reaction lives with that trade, and the usual answer is a temperature low enough for a decent balance with a catalyst good enough to reach it.

TAKEAWAY 17W

One setting moving two quantities in opposite directions is the normal case rather than a special one.

END OF THE SOL 14W

A plant is a chain, and the fastest step decides nothing on its own.

The loop that pays for itself

PLAN CARD 124W

Sol 294. The exchanger was warmed out overnight and the cold end took liquid again at six hours, which leaves the plant with the question it had two sols ago and better information about it. Herrera's loop can convert two thirds of a pass and recycle the rest; what it cannot do is make methane out of carbon dioxide it has not been given, or hydrogen the stacks have not made. Today you size the feed the bed needs at the new rate, find out why Achebe's stacks return less hydrogen than the current says they should, and look at what the water carries with it. Abiola has been asking for a second polishing column since sol 240 and has not been given 1.

BRIEFING 17W

With the cold line clear, the question is whether the feed can keep up with the bed.

WHAT YOU DECIDE 13W

Size the feed the plant needs, and find where the current is going.

4.1 How much gas the bed has to be fed

REACTOR HALL • A MODULE • BALLPARK

SCENE 31W

The plant has to average 13.8 kilograms of methane a sol from here to the window. Herrera wants the carbon dioxide that implies before the intake compressors are asked for anything.

GUIDE 59W

Five numbers, and two of them belong to other parts of the equation: the hydrogen ratio, and the water produced. Ask of each whether this carbon bookkeeping needs it. One carbon atom goes from each carbon dioxide into one methane. So the mole ratio is one to one and the mass ratio is not, because the molar masses differ.

ASK 11W

Estimate the carbon dioxide the reactor must be fed each sol.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

one mole of carbon dioxide yields one mole of methane in this reaction • a molar mass converts moles to kilograms

VERDICT 76W

Stoichiometry is mole bookkeeping. The balanced reaction sends one carbon atom from each carbon-dioxide molecule into one methane molecule. Mass adds one conversion step. Carbon dioxide has a molar mass of 44 g/mol, while methane is 16 g/mol. Each kilogram of methane therefore needs 44/16 kilograms of carbon dioxide. The intake compressors must gather that feed from a very thin atmosphere. The required compressor work scales directly with the production target the plant promises to meet.

TAKEAWAY 20W

Stoichiometry turns a production target into a feed rate, and the feed rate is what the intake has to meet.

4.2 Where 8% of the charge went

ELECTROLYSIS HALL · A MODULE · BALANCE

SCENE 35W

The cell held 640 amps for a full sol. Faraday's law predicts about 594 grams of hydrogen, but the separator collected only 546. Achebe has already checked the collection-line pressure and the hydrogen inventory downstream.

GUIDE 56W

Read every stream, then count only what belongs in the charge ledger. Faraday's law says 640 amps for a sol should give about 594 grams of hydrogen and the separator collected 546. Achebe has already cleared the collection line and the downstream inventory, so the missing 8% is a term the ledger does not yet have.

ASK 15W

What term closes the charge ledger, and what does that term measure about the cell?

BEHIND THE BUTTON 131W

What Faraday's law is claiming. Every electron that crosses the cell makes a fixed amount of hydrogen. So the predicted mass is arithmetic on the charge, and it is an upper bound: it assumes every electron did the intended reaction.

Why the gap is a measurement rather than an error. Electrons that did something else — a side reaction, a shuttle current, recombination inside the cell — still crossed and still counted as charge. The shortfall is therefore a measurement of how much of the current did useful work, which is the cell's current efficiency.

Why checking the line first was right. A leak or unmeasured inventory downstream would explain the gap without saying anything about the cell. Ruling those out is what turns a discrepancy into evidence about the electrochemistry.

TAKES AS READ

Faraday's law converts charge into a theoretical amount of product · current efficiency is collected product divided by theoretical product from charge

VERDICT 79W

Current times time gives charge, and Faraday's law converts that charge into a theoretical hydrogen mass. For this run the charge corresponds to about 594 grams. Only 546 grams were collected, while line inventory and measured vent loss were unchanged. The missing 48 grams are therefore not a hidden hydrogen store. They are hydrogen-equivalent charge that crossed by other electrochemical paths or recombined before collection. The ratio 546/594 is about 92%, which is the current efficiency for this run.

TAKEAWAY 20W

Faraday's law fixes the theoretical product from charge; current efficiency measures how much of that charge reached the intended reaction.

4.3 What comes with the water

WATER PLANT · A MODULE · PROTOCOL

SCENE 35W

The holding tank's analysis is on Abiola's bench: four dissolved species, none of them water. Each one has somewhere downstream that it matters, and the plant has one polishing column between them and everything else.

GUIDE 48W

Four dissolved species and four consequences. Pair them by asking what surface or process each one reaches downstream. One binds to nickel. One deposits on hot faces. One fouls mechanically. And one is already handled as a process gas, which is why not every impurity is equally dangerous.

ASK 9W

Match each dissolved species to what it does downstream.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

salts dissolve into ions in water · an electrolysis cell and a catalyst bed both have surfaces that can be fouled · intermolecular forces, and why they set a boiling point — taken as read

VERDICT 76W

A dissolved species matters because of the surface or process it reaches downstream. Chloride is dangerous here because it can bind strongly to nickel catalyst sites. Sulphate can contribute to scale on hot surfaces. Fine particles create a mechanical fouling problem. Dissolved carbon dioxide is different because the plant already handles carbon dioxide as a process species. The polishing train should therefore be designed around consequences, not around the idea that every impurity is equally harmful.

TAKEAWAY 14W

Water from the ground is a solution, and every solute has an address downstream.

4.4 Three hundred degrees, while the feed wanders

REACTOR HALL · A PERSON · HOLD

SCENE 36W

The Sabatier bed has to sit at its working temperature while the electrolysis stacks feed it. The stacks follow the array, the array follows the sun, and the bed heats itself whenever it is making methane.

GUIDE 53W

Hold the bed at its working temperature inside the band on the panel. The band narrows as the sol goes on, because the tank has to be filled at a stated rate and a bed wandering by ten degrees is not making the rate anybody planned. The cooling loop valve is your control.

ASK 10W

Hold the bed at its working temperature through the sol.

BEHIND THE BUTTON 92W

Why the bed heats itself. Assembling methane gives out heat, so a bed that is working is a bed that is warming, and a bed that warms converts less per pass because the ceiling falls with temperature. The reaction and the cooling are in a loop with each other.

Why the feed will not sit still. The stacks run on array power and the array follows the sun and the dust. More hydrogen arriving is more reaction, which is more heat, and it arrives as a rate rather than as a step.

TAKES AS READ

a reaction that gives out heat warms what it is running in

VERDICT 149W

The Sabatier reaction gives out heat, so the bed warms whenever it is working, and a warmer bed has a lower equilibrium ceiling — which is the whole reason this plant cares about temperature rather than treating it as a comfort setting. Every disturbance here is a rate. The stacks pushing more hydrogen do not raise the bed by five degrees and stop; they raise the reaction rate, which raises the heat output, which goes on raising the bed until the cooling is set to carry it away. So the valve is moved to a new position and left there. The band narrows because the tank has a filling schedule and the ascent vehicle has a launch window: a bed wandering ten degrees either way is not making the rate the plan was built on, and by late sol there is no margin left to make it up in.

TAKEAWAY 13W

A process that heats itself is held by whoever is watching the cooling.

4.5 What freezes before methane condenses — Review

COLD END · A MODULE · CALLBACK · SEQUENCE

SCENE 30W

A different stream comes off the greenhouse: mostly nitrogen, with water vapour, some carbon dioxide and a little argon. It is being cooled to recover the argon as a liquid.

GUIDE 58W

All four substances are in the stream together, so nothing here happens before anything else. What orders them is how strongly their molecules hold on to one another. Ask of each what kind of attraction it has: hydrogen bonds, weaker forces between molecules, or only the fleeting pull between electron clouds. One of them is a single atom.

ASK 14W

Order the substances by the temperature at which they leave the gas, warmest first.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

each substance leaves the gas phase at its own temperature · stronger attractions between molecules mean a higher condensation temperature

VERDICT 81W

The order is the strength of the forces between whole molecules. Water hydrogen-bonds and is the first out by a wide margin. Carbon dioxide is held by weaker forces and, at this pressure, goes straight from gas to solid. Nitrogen and argon are held only by the fleeting attractions between electron clouds, so they need the deepest cold, and argon — one atom, with the least to hold onto — is last. A cooling stream is a separation in that order.

TAKEAWAY 14W

Cooling a mixture separates it in the order the attractions between its molecules run.

END OF THE SOL 20W

A plant is limited by whichever feed runs out first, and it will not be the same one every sol.

The ceiling on running hotter

PLAN CARD 122W

Sol 295, and Sundqvist has asked for another 40 degrees on the bed, 4 sols after the first 40 worked. Kaur will not sign for it: the last charge she pulled had grown its metal particles, and a bed that has sintered does not come back. Demir wants to know where the heat is coming from, since the reactor gives out heat and the habitat is short of it. Today you put an arithmetic ceiling on the temperature argument, decide what a catalyst can and cannot change about it, and follow the product stream into the cold end. The number you produce settles a fortnight of argument in either direction, and Wells has said she will act on it the same evening.

BRIEFING 16W

Sundqvist has asked for another 40 degrees, and this time the question has an arithmetic answer.

WHAT YOU DECIDE 12W

Put a number on how far the temperature argument can be taken.

5.1 The temperature the reaction stops wanting to run

PLANT CONTROL • A MODULE • BALLPARK

SCENE 32W

Demir has thermodynamic data for the Sabatier reaction on the desk. The sheet lists its enthalpy and entropy changes per mole of methane. He wants ΔG evaluated at the present bed temperature.

GUIDE 52W

Five numbers, and two of them belong elsewhere: the gas constant, and room temperature. Ask of each whether this calculation needs it. The two terms oppose each other, and the temperature multiplies only one of them. So the balance between them shifts as the bed gets hotter, and somewhere it changes sign.

ASK 11W

Estimate the free-energy change for the reaction at the bed's temperature.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

free energy combines the heat of a reaction with its entropy change • a negative free-energy change means the reaction can run on its own

VERDICT 81W

The enthalpy and entropy terms oppose each other. Methanation releases heat, which favours products, but it also reduces the number of gas molecules and gives a negative entropy change. The $T\Delta S$ term therefore becomes more important as temperature rises. At 593 K the estimate is still strongly negative. Using the same tabulated ΔH and ΔS gives 0 crossing near 960 K. Real thermodynamic values vary somewhat with temperature, so 960 K is a course-model threshold rather than an exact plant limit.

TAKEAWAY 18W

With fixed tabulated ΔH and ΔS , $\Delta G = \Delta H - T\Delta S$ predicts the temperature where spontaneity changes sign.

5.2 What a catalyst cannot do

CATALYST BAY · A PERSON · CHOICE

SCENE 36W

Kaur has two charges on the bench: the spent 1 from sol 288 and a fresh 1 still in its sealed can. Sundqvist's proposal assumes a better catalyst would raise the methane a pass can give.

GUIDE 63W

Four accounts of what a better catalyst buys. Ask of each whether the catalyst could deliver it. It lowers the barrier by the same amount in both directions, so their ratio is untouched, and that ratio is the equilibrium constant. Nor does the route change how much heat comes out. What it buys is time, which a fixed number of sols makes valuable.

ASK 13W

A better catalyst is fitted at the same temperature and pressure. What changes?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a catalyst offers a reaction an easier path · equilibrium is where the forward and reverse rates match

VERDICT 89W

A catalyst lowers the activation barrier, and it lowers it by the same amount in both directions. Forward and reverse both speed up, their ratio is untouched, and that ratio is the equilibrium constant. Nor does the catalyst change how much heat the reaction gives out: enthalpy depends on what the molecules are, not on the road taken between them. So a better catalyst buys time, which in a plant with a fixed number of sols is worth a great deal — and it buys no yield at all.

TAKEAWAY 17W

A catalyst changes how long the balance takes to arrive and leaves the balance where it was.

5.3 The order the cold end has to work in

COLD END · A MODULE · SEQUENCE

SCENE 36W

Cho is rebuilding the cold end's written procedure after the blockage, and wants the four stages back in the order the stream meets them. 2 of the 4 exist because of what happened on sol 293.

GUIDE 55W

All four stages exist in the plant, so the order is about what each removes before the next. Ask of each what would foul the colder stage after it. Water freezing in an exchanger. Carbon dioxide depositing as a solid. And compression is different in kind: it raises the temperature at which everything below condenses.

ASK 10W

Order the stages of the cold end, warmest to coldest.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a substance condenses at a temperature that rises with pressure · carbon dioxide turns to solid near 195 K at these pressures · vapour pressure, and reading a cryogenic tank by it — taken as read

VERDICT 82W

The stream is cleaned as temperature falls. Each stage sits at a temperature where it can remove one unwanted species before that species harms a colder stage.

Compression raises condensation temperatures and can reduce refrigeration demand. Water is removed before it can freeze in the exchanger. Carbon dioxide is trapped at a warmer stage where deposited solid can be managed. Only the cleaned remainder goes to the methane-condensation stage near 112 K. Driving contaminated gas colder wastes energy and creates blockage risk.

TAKEAWAY 13W

Cleaning a stream is cheapest at the warmest temperature that will do it.

5.4 What the loop settles at — Review

REACTOR HALL · A MODULE · CALLBACK · PROTOCOL

SCENE 33W

A second reactor on the station makes ammonia for the greenhouse, from nitrogen and hydrogen, and it also gives out heat and ends with fewer gas molecules. Four changes are proposed to it.

GUIDE 54W

Four changes and four effects, in a reactor making something else entirely. Pair them the same way. Count the gas molecules on each side for pressure. Ask which side the heat is on for temperature. Ask whether anything taking part was added or removed. And a catalyst changes the time rather than the destination.

ASK 12W

Match each change to what it does to the ammonia at equilibrium.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a reaction that gives out heat has its balance pushed back by heating · fewer gas molecules on one side makes that side favoured by pressure

VERDICT 70W

The shape of the equation decides the answers, which is why a reactor making a completely different substance behaves the same way. Count the gas molecules to know what pressure does. Ask which side the heat is on to know what temperature does. Ask whether anything taking part has been added or removed to know whether the balance moves at all. A catalyst changes the time and never the destination.

TAKEAWAY 16W

A rule about balance is about the equation's shape, not about the substance in the pipe.

END OF THE SOL 13W

An argument that has an arithmetic answer should not be settled by seniority.

Water is the whole plant

BEFORE THE SOL — HUNT

Six regolith bins, and the tally says four

The feedstock tally is what the whole propellant schedule is planned against, and two bins are not where it says they are. A bin nobody can find is either full and idle or empty and counted, and those two mean very different launch dates.

PLAN CARD 126W

Sol 296, and for the third sol running the plant has made about 11 kilograms with a bed that could do more. Sora Nakagawa, who cuts the ground the water comes from, has been working poorer ice for a week: 8% water by mass in the spring, four and a half this morning, and the machine cannot go deeper fast enough to make it up. Abiola's holding tank is brine rather than water, and it stays liquid at temperatures that ought to have frozen it solid. Today you find out how salty that has to be to behave that way, what the stacks do with the voltage above the minimum, and which way the loop is running right now. Everything downstream is waiting on one tank.

BRIEFING 16W

The bed is fine and the stacks are fine, and the plant made 11 kilograms again.

WHAT YOU DECIDE 14W

Find the feed that actually caps the plant, and what it costs to clean.

6.1 The tank that will not freeze

WATER PLANT · A MODULE · BALLPARK

SCENE 35W

The holding tank sat at minus 40 overnight and drained as a liquid this morning. Abiola has the analysis: magnesium perchlorate, and a great deal of it. He wants the freezing point that concentration implies.

GUIDE 58W

Five numbers, and two of them belong to other questions: the boiling-point constant, and a molar mass. Ask of each whether this shift depends on it. What matters is how many particles a formula unit releases, not how heavy it is. And note the limit: at eight molal the ideal answer is a magnitude rather than a value.

ASK 13W

Estimate how far this brine's freezing point falls below that of pure water.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a dissolved salt separates into ions in water · the freezing point of pure water is 273 K

VERDICT 89W

Dissolved particles lower the chemical potential of liquid water and make freezing occur at a lower temperature. In the ideal colligative model, the size of that shift depends on the number of dissolved particles. Magnesium perchlorate contributes three ions per formula unit, so the van 't Hoff factor is three in the course calculation. At 8 molal the solution is extremely concentrated, so real ion interactions make the ideal 45 K estimate only a rough magnitude. The calculation still shows why this brine remains liquid far below 0 °C.

TAKEAWAY 18W

In the ideal colligative model, freezing-point depression scales with the number of dissolved particles, not their chemical identity.

6.2 What the extra volt is doing

ELECTROLYSIS HALL • A PERSON • CHOICE

SCENE 38W

The water stack needs 1.23 volts a cell in principle and Achebe runs it at 1.75. The difference is half a kilowatt across the stack, and Demir wants to know whether that energy ends up in the hydrogen.

GUIDE 56W

Four accounts of where the extra half-volt goes. Ask of each whether it changes the product or the cost. The minimum voltage pays for the free-energy change. Anything above it is the price of driving current through real electrodes. Hydrogen made at 1.75 volts is the same hydrogen, and the difference has to end up somewhere.

ASK 12W

Where does the energy from the extra half-volt a cell end up?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a cell has a minimum voltage set by the thermodynamics of the reaction • electrical power is current times voltage

VERDICT 82W

The thermodynamic minimum voltage pays for the free-energy increase of splitting water. Any voltage above that minimum is overpotential. It is the extra cost of driving useful current through real electrodes and electrolyte. Part of that cost comes from electrical resistance. Part comes from slow electrode steps that need additional driving force. The extra electrical energy ends up mainly as heat in the stack.

Hydrogen made at 1.75 V is chemically the same product as hydrogen made closer to the reversible voltage.

TAKEAWAY 19W

A cell run fast costs more per mole than a cell run slowly, and the difference leaves as heat.

6.3 Compute which way it runs

REACTOR HALL · A MODULE · BALLPARK

SCENE 32W

Herrera freezes the 09:40 gas-analyser screen before the loop has settled. 4 partial pressures sit beside the equilibrium constant for 593 K. He wants the direction called before anyone changes a valve.

GUIDE 57W

Six numbers, and two of them are not partial pressures: the total loop pressure, and the constant the answer is compared against. Ask of each whether the quotient uses it. The expression is the same shape as the constant, applied to the mixture that exists now. And note what the comparison gives: a direction, not a speed.

ASK 14W

Estimate Q from the four partial pressures. Is the mixture driven forward or backward?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

for $\text{CO}_2 + 4\text{H}_2 \rightleftharpoons \text{CH}_4 + 2\text{H}_2\text{O}$, $Q_p = P_{\text{CH}_4}(P_{\text{H}_2\text{O}})^2 / [P_{\text{CO}_2}(P_{\text{H}_2})^4] \cdot K_p$ is 0.14 at this bed temperature

VERDICT 77W

The reaction quotient uses the same pressure expression as the equilibrium constant, but it uses the mixture that exists right now. Here Q is about 0.016 while K is 0.14. The mixture therefore has too little product relative to equilibrium, so the forward reaction is favoured until Q rises to K . The comparison predicts direction, not speed. A poor catalyst could take a long time to close the gap. Temperature can change K itself; the catalyst cannot.

TAKEAWAY 19W

$Q < K$ drives the reaction forward, $Q > K$ drives it backward, and $Q = K$ is equilibrium.

6.4 How long a molecule stays in the bed

CATALYST BAY • A MODULE • CALLBACK • BALLPARK

SCENE 35W

Kaur wants the contact time worked out at the new flow before anything else is changed. The bed holds 340 litres of packed catalyst and the loop is passing 85 litres a minute through it.

GUIDE 60W

Five numbers, and three of them describe the conditions rather than the geometry: the pressure, the temperature, and the feed ratio. Ask of each whether this time depends on it. Residence time is how long the gas stays with the catalyst. Push more gas through the same bed and that contact shortens, so flow and conversion move against each other.

ASK 11W

Estimate how long a molecule of gas spends inside the bed.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a flow quoted in litres a minute is a volume passing each minute • gas volumes depend on the pressure and temperature they are quoted at • rate constants, and how steeply they climb with temperature — taken as read • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 75W

Residence time is the contact the reaction gets. A rate constant says how quickly reaction occurs at the stated conditions. Residence time says how long the gas remains with the catalyst. Conversion depends on both. Raising throughput by pushing more gas through the same bed shortens that contact. The plant can therefore increase flow while lowering conversion in one move. Rate and residence time must be judged together before a higher production setting is accepted.

TAKEAWAY 18W

Conversion depends on how fast the reaction runs and on how long it is allowed to run for.

END OF THE SOL 14W

The step that limits a plant is not the step anybody is arguing about.

The bed walks downstream

PLAN CARD 140W

Sol 297, and Kaur brought in 9 sols of temperature profiles this morning with the hot spot moved 30 centimetres down the bed. Output is falling about 2% a sol against a feed that has not changed, and the assay bench has begun reporting carbon monoxide in the product where there was none in the spring. Élodie Moreau, who works the transfer lines, has a boil-off log that has gone up as well, which may be the same problem or a different one. Today you decide what the bed is telling you, work out why a hot reactor makes a gas nobody asked for, and put a number on what the tank farm loses standing still. Kaur says the profile means the front of the bed is dead; Sundqvist says a bed that hot is doing what hot beds do.

BRIEFING 14W

The reactor's hot spot has moved 30 centimetres down the bed in 9 sols.

WHAT YOU DECIDE 12W

Find out what is happening inside a bed nobody can see into.

7.1 What the tank farm loses standing still

COLD END · A MODULE · BALLPARK

SCENE 35W

Moreau's log shows the oxygen tank losing liquid on sols when nothing is drawn from it. The tank's insulation lets about 30 watts through, and she wants that expressed in kilograms rather than in watts.

GUIDE 50W

Five numbers, and two belong to other questions: the boiling point, and the molar mass. Ask of each whether this loss depends on it. A liquid sitting at its boiling point spends incoming heat on vaporising rather than warming. Thirty watts sounds small and acts every second of the sol.

ASK 10W

Estimate the oxygen the tank loses to boil-off each sol.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a boiling liquid takes in heat without rising in temperature · a sol is about 88,800 seconds long · vapour pressure, and reading a cryogenic tank by it — taken as read

VERDICT 80W

A cryogenic liquid sits near its boiling point. Heat that enters is therefore spent mainly on vaporising liquid rather than raising its temperature. The required energy per kilogram is the latent heat of vaporisation. That turns an insulation leak directly into a propellant-loss rate. 30 watts sounds small, but it acts every second of the sol. Over 88,800 seconds it boils away about 12.5 kg of oxygen. Repeated across many sols, a modest heat leak becomes a major production loss.

TAKEAWAY 14W

Heat reaching a cryogenic tank is paid for in kilograms rather than in degrees.

7.2 The reaction that likes being hot

PLANT CONTROL • A PERSON • CHOICE

SCENE 43W

Dana Stern, the analytical chemist, has carbon monoxide rising in the assay each time the reactor set point is raised. Methane falls by almost the same carbon amount. Kaur has the two candidate reactions on the board and wants the temperature effect explained.

GUIDE 58W

Four accounts of why the hotter bed makes more carbon monoxide. Ask of each whether it treats the two reactions separately. One releases heat and one takes it in, so their constants move in opposite directions when the bed is warmed. Both may also get faster. What changes is not only conversion but how the carbon is divided.

ASK 10W

Why does the hotter bed make relatively more carbon monoxide?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

free energy combines an enthalpy term with a temperature times an entropy term • a reaction that takes in heat has a positive enthalpy change • rate constants, and how steeply they climb with temperature — taken as read

VERDICT 76W

Competing reactions must be considered separately. The Sabatier reaction is exothermic, so heating shifts its equilibrium toward reactants. The carbon-monoxide-forming side reaction is endothermic, so heating shifts its equilibrium toward products. Both reactions may also become faster as temperature rises. The important point is selectivity: the two equilibrium constants do not move in the

same direction. A temperature set point therefore changes not only total conversion, but also how the carbon feed is divided among products.

TAKEAWAY 14W

Temperature can change selectivity because competing reactions can have different thermodynamic responses to heating.

7.3 Reading a bed by where its heat is

CATALYST BAY · A MODULE · DIAGNOSIS

SCENE 39W

Nine sols of profiles are pinned along the wall in order. The hot spot starts 4 centimetres into the bed on the first and sits at 34 on the last, with about the same peak temperature on every 1.

GUIDE 63W

Five readings, and the unchanged ones do the work. The peak temperature is the same, so wherever the reaction happens it is going as hard as ever. Pressure drop is unchanged, which rules out packing and fouling. The feed is unchanged. Ask of each candidate how many of the five it fits, and note which end of a bed sees the strongest feed.

ASK 5W

Which explanation fits every reading?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a reaction gives out its heat where it is happening · a bed can lose activity in one part and keep it in another

VERDICT 82W

An unchanged peak temperature is the reading that decides it: wherever the reaction is happening it is going as hard as ever, so the catalyst that still works has not been weakened. What has changed is how much of the bed is doing it. Unchanged pressure drop rules out packing and fouling; unchanged feed rules out the third explanation. A bed dies from the inlet end, since that is where the feed is strongest, and conversion falls as the working length shrinks.

TAKEAWAY 11W

Where a bed is hot is where it is still working.

7.4 Which electrode does which job — Review

ELECTROLYSIS HALL • A MODULE • CALLBACK • PROTOCOL

SCENE 29W

The station's oxygen generator splits water in a low-temperature cell rather than a hot one. Achebe wants it read the same way as the hot stacks, part by part.

GUIDE 57W

Four parts of a cell and four things that happen. Pair them by asking which way the electrons are pushed. In means reduction, and hydrogen appears there in any water-splitting cell, hot or cold. Out means oxidation. The separator carries one ion and keeps the gases apart. And one item is a threshold rather than a place.

ASK 10W

Match each part of this cell to what happens there.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

reduction happens where electrons are pushed in • the electrolyte carries ions and not electrons

VERDICT 75W

A cell is a redox reaction with the halves separated, and which half sits where is fixed by the direction the electrons are pushed. Electrons in means reduction, so hydrogen appears at that electrode in any water-splitting cell, hot or cold. Electrons out means oxidation and oxygen. The separator carries one ion and stops the gases mixing. Below the minimum voltage nothing happens, and above it the excess is the price of a useful rate.

TAKEAWAY 16W

The parts of an electrolysis cell do the same jobs whatever the cell is made of.

END OF THE SOL 16W

A measurement taken every sol becomes evidence when somebody lays nine of them side by side.

The assay bench

PLAN CARD 128W

Sol 298, and Dana Stern, the analytical chemist, has run the first full product assay since the spring rather than the two-line check the plant has been living on. Four numbers came off it, and one is a compound nobody had been asking about, which means it has been in the stream for as long as nobody asked. Achebe wants her stack's output stated in kilograms rather than in amps for the same reason. Today you convert a sol of current into a mass of hydrogen, read the assay for what each line implies about the loop, and look at how the polishing column behaves as it fills. Anaya Bhatt, the ascent vehicle systems engineer, has asked for all three before she countersigns another batch into her tanks.

BRIEFING 17W

Stern has run a full assay on the product stream for the first time since the spring.

WHAT YOU DECIDE 13W

Turn current into kilograms, and find out what is actually in the product.

8.1 From amps to kilograms

ELECTROLYSIS HALL · A MODULE · BALLPARK

SCENE 32W

1 cell in the water stack carried 640 amps for the whole sol. Achebe wants the hydrogen that represents, worked out from the charge rather than from what the collection line reported.

GUIDE 44W

Five numbers, and one of them is the cell voltage, which belongs to the energy bill rather than to the amount. Ask of each whether this mass depends on it. Charge divided by the charge per mole gives moles. And note what the answer

ASK 10W

Estimate the hydrogen one cell should produce over a sol.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

1 mole of electrons carries 96,485 coulombs · splitting water to hydrogen takes two electrons per molecule · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 76W

Faraday's law is charge bookkeeping. Current multiplied by time gives total charge. Dividing by the charge required per mole gives moles of hydrogen. Multiplying by hydrogen's molar mass gives the product mass. Voltage does not enter this amount calculation; voltage controls the energy cost. The result is a theoretical maximum because it assumes every electron contributes to the intended electrode reaction. Comparing this number with collected hydrogen later gives the current efficiency of the real cell.

TAKEAWAY 16W

Charge and moles are one measurement in two units, once the electrons per molecule are known.

8.2 Four lines off the assay bench

REACTOR HALL • A PERSON • PROTOCOL

SCENE 31W

Stern's assay of the product stream has four lines on it. Herrera wants each one matched to what it implies about the loop before anybody argues about a set point again.

GUIDE 59W

Four assay lines and four meanings. Pair them by asking which part of the plant each line reports on. The headline figure is the least informative. One line is evidence of a second reaction on the same bed. One is the recycle working as designed. And one is a maintenance item that becomes an emergency two hundred metres downstream.

ASK 11W

Match each assay line to what it says about the loop.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

the loop returns unreacted gas to the reactor inlet • the product stream is dried before it is cooled • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 81W

Every line in an assay belongs to a part of the plant. The methane figure is the headline and the least informative. Carbon monoxide is not a contaminant that blew in; it is evidence of a second reaction on the same bed. Unreacted hydrogen is expected in a loop that recycles and would be alarming in one that did not. Water past the drier is a maintenance item that becomes an emergency 200 metres downstream, at 273 K in the exchanger.

TAKEAWAY 16W

An essay is a statement about the process, not only about the bottle it came from.

8.3 The bed that warns only at the outlet

WATER PLANT · A MODULE · CHOICE

SCENE 42W

The polishing column is a packed bed of ion-exchange resin with a fixed number of sites in it. Abiola has been asked how the plant would know the column was full, and he says the honest answer is that it would not.

GUIDE 64W

Four accounts of what the outlet does. Ask of each whether it fits a bed with a finite number of sites. The sites fill from the inlet end, so a front moves through the resin. Two of these options have the outlet degrading or improving gradually, which is not what a moving front produces. And the outlet is the only place the warning appears.

ASK 15W

What happens at the outlet as the exchange front reaches the end of the bed?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an ion-exchange bed swaps unwanted ions for harmless ones · a packed bed holds a finite number of ions · rate constants, and how steeply they climb with temperature — taken as read · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 84W

An ion-exchange bed has a finite number of exchange sites. A loading front moves through the resin as those sites fill. The outlet can stay very clean for most of the run, then rise quickly when that front reaches the end. The transition is called breakthrough. It is not perfectly instantaneous, and the exact curve depends on flow and mass transfer. The practical warning is therefore an outlet measurement.

Monitoring conductivity often enough lets the plant act before the effluent reaches the inlet concentration.

TAKEAWAY 15W

A bed with a capacity has two states, and there is no announcement between them.

8.4 What has to be paid for — Review

PLANT CONTROL • A MODULE • CALLBACK • CASEBOOK

SCENE 32W

Demir has four of the station's processes on the board and wants them sorted before the quarterly energy plan is written. Two of them run downhill and two have to be pushed.

GUIDE 56W

Four processes and four demands. Pair them by asking two things of each: does it run downhill or uphill, and is the bill chemical or physical? Only direction decides whether there is a bill at all. Boiling runs downhill, which is why stopping it costs work. And compression is uphill and paid for at the compressor.

ASK 10W

Match each process to what it needs from the station.

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a negative free-energy change means a process can run on its own • running on its own says nothing about how fast a process goes

VERDICT 77W

Direction and speed are separate questions, and only direction decides whether there is a bill. The assembly is downhill and slow, which is what a catalyst is for. Splitting water is the same reaction backwards and the bill is the free-energy difference, paid in electricity. Boiling is downhill, which is why a tank needs work done on it to stop happening. Compression is uphill and physical rather than chemical, and it is paid for at the compressor.

TAKEAWAY 19W

Sorting a station's processes by direction is what says which of them have a bill nobody can design away.

END OF THE SOL 14W

A measurement nobody asks for is a measurement that arrives later at full size.

What the spent charge was carrying

BEFORE THE SOL — CANVASS

Who has actually seen the seal on module four

The habitat's leak rate went up on sol 9 and the log blames a seal nobody has inspected since landing. Ask round the crew until enough of them have said what they have actually seen rather than what the log says, because six people is a small enough sample that one assumption becomes a fact.

PLAN CARD 124W

Sol 299, and the spent charge Kaur pulled on sol 288 came back from Stern's bench an hour ago. It had sat in a sealed can for 11 sols, because nobody thought a dead catalyst would say anything a temperature profile had not. The result is one line long and it changes what the last fortnight was about. Abiola's outlet conductivity trace has been printed every sol since the plant started and nobody has had a reason to read it. Today you work out what killed the bed, when it started, and what the loop's poor conversion has really been telling everybody. Sundqvist and Herrera have argued about a set point for 8 sols, and 11 sols of production are gone while they did.

BRIEFING 12W

The spent catalyst charge came back from the assay bench this morning.

WHAT YOU DECIDE 13W

Find out what has been taking the bed apart, and for how long.

9.1 One line off the spent charge

CATALYST BAY • A MODULE • CHOICE

SCENE 35W

The spent charge was assayed in three sections: the first 20 millimetres, the middle, and the outlet end. Kaur has the five findings on the board and is no longer sure of her own explanation.

GUIDE 58W

Four explanations for a dead front section. Ask of each what the material evidence would look like. Heat-driven coarsening changes particle size, and the particles match fresh material. Heavy coking leaves carbon, and only traces are there. A faulty charge would be weak throughout, and the outlet still works. So the pattern is something arriving with the feed.

ASK 5W

Which explanation fits every finding?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a catalyst can lose activity by heat or by a substance binding to it • heat makes small metal particles merge into larger ones

VERDICT 81W

One material observation settles the main dispute: the nickel particles are still the same size as fresh material. That argues against heat-driven sintering. Trace carbon also makes heavy coking unlikely. A uniformly defective charge should be weak throughout its length, but the outlet section remains active. The remaining pattern fits something arriving with the feed and binding first near the inlet. That mechanism also explains why the active hot zone moved downstream as more of the front section lost usable surface.

TAKEAWAY 17W

Two causes with one symptom are told apart in the material rather than in the plant readings.

9.2 The trace nobody read

WATER PLANT · A PERSON · CHOICE

SCENE 35W

The conductivity of the water leaving the polishing column is logged every sol. The trace runs flat for 260 sols, steps up over about 4, and runs flat and high from there to this morning.

GUIDE 59W

Four readings of one trace, and the shape carries the history. Ask of each what the inlet record would have to show. A saltier feed would appear there, and it is flat. A sensor reset would give a step rather than a rise. What a filling bed produces is low, then rising toward the inlet value, then staying high.

ASK 8W

What does the shape of that trace say?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

conductivity rises with the ions dissolved in water · an exchange bed holds a fixed number of ions · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 82W

The shape of the record carries the history. A fresh exchange bed keeps the outlet conductivity low while unused sites remain. As the loading front reaches the outlet, conductivity rises toward the inlet value and then stays high. That is the breakthrough signature. A saltier feed would also appear in the inlet record, but the inlet stays flat. A sensor reset would produce a much sharper discontinuity. The trace therefore dates breakthrough to about sol 261 and shows prolonged operation after it.

TAKEAWAY 17W

A step in a record is a date, and a date turns a fault into a history.

9.3 Change the time, not the ceiling

REACTOR HALL · A MODULE · CONTROL

SCENE 33W

Herrera and Sundqvist have argued for a fortnight about whether the hot reactor is at equilibrium or simply short of reaction time. Stern gives them one controlled run before the production schedule resumes.

GUIDE 74W

The number you are watching is conversion — how much of the gas that went in came out as product. Three settings can be changed, and only one of them separates the two explanations on the table: too little time in the reactor, or a ceiling set by the equilibrium itself. A response counts only if it beats the assay's own scatter. Change one, run it, restore it, and name the setting conversion follows.

ASK 18W

Which controlled change separates a kinetic limit from an equilibrium limit, and does conversion follow it on reversal?

BEHIND THE BUTTON 157W

What the argument is about. A reaction that has not finished and a reaction that has reached its limit both give conversion below the ideal, and they read the same on a single measurement. The difference is what they respond to: more time helps the first and does nothing for the second.

What each setting does. Gas flow sets how long the gas spends over the bed, so it changes the time available without changing what the reaction is allowed to reach. Temperature and pressure both move the equilibrium ceiling itself. That is why one of the three answers this question and the others answer a different one.

Why the reversal matters on a plant. A bed that is ageing, a heater drifting, a feed composition changing — all of them move conversion slowly while you work. Putting the setting back and watching conversion return is how you tell your own change from the plant's own drift.

TAKES AS READ

equilibrium is a ceiling that depends on temperature and pressure • conversion also depends on how long the gas spends in the bed • rate constants, and how steeply they climb with temperature — taken as read • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 84W

Equilibrium at fixed temperature and pressure does not depend on how quickly gas crosses the bed. Kinetics does. Halving the flow doubles residence time while leaving the equilibrium composition unchanged. In this controlled run, conversion rises well beyond repeat scatter and returns when the original flow is restored. That means the reactor had been leaving before the mixture reached its equilibrium ceiling. Changing temperature or pressure would not answer the same question because each would move the equilibrium as well as the reaction rate.

TAKEAWAY 18W

Changing residence time at fixed temperature and pressure changes kinetic approach to equilibrium without moving the equilibrium ceiling.

9.4 Reading a blockage from outside the pipe — Review

COLD END · A MODULE · CALLBACK · DIAGNOSIS

SCENE 34W

A line on the oxygen side has slowed over two sols. Cho has four readings from around it, and the plant needs to know whether to warm the line out or open it up.

GUIDE 58W

Five readings, and three of them rule things out. A cold head at its set point. A steady jacket pressure. A flow falling over two sols rather than in a minute. Ask of each candidate how many of the five it fits. And note why the material matters: one of them clears itself when the line is warmed.

ASK 5W

Which explanation fits every reading?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a refrigerator at its set point is doing its job · carbon dioxide deposits as a solid near 195 K

VERDICT 80W

Three readings do the ruling out. A cold head at its set point has not lost capacity, and a steady jacket pressure is intact insulation. A valve moving would change the flow in the minute it moved rather than over two sols. What is left is something accumulating, and the trap's breakthrough figure names the material. It matters which material: carbon dioxide sublimates away when the line is warmed to 220 K, so this 1 is cleared without opening anything.

TAKEAWAY 13W

What a blockage is made of decides whether warming the line clears it.

END OF THE SOL 12W

Two people can argue correctly for a fortnight about the wrong variable.

A new charge and a rule for the water

PLAN CARD 132W

Sol 300. The last spare catalyst charge came out of stores this morning and there is no second one on this planet, so whatever is done with it is done once. Kaur wants it qualified properly before it carries production; Sundqvist wants it making methane by the afternoon, and this time Marion Wells, who commands the station, has come down on Kaur's side. Achebe has parts enough for one improvement to the water and power loop and a list of four. Today you set the order the charge is brought into service, decide what the spare parts are spent on, and settle whether the methane tank's rising pressure is a leak or a temperature. Everything the plant does for the next hundred and 80 sols runs through a bed commissioned this afternoon.

BRIEFING 14W

There is one spare catalyst charge on the station and no second one coming.

WHAT YOU DECIDE 13W

Get a bed back into service and decide what stops this happening again.

10.1 Bringing a charge into service

CATALYST BAY · A MODULE · SEQUENCE

SCENE 38W

The fresh charge arrives as nickel oxide on alumina, which is not a catalyst yet. Kaur has four operations written out and wants them in an order she can hand to a technician at three in the morning.

GUIDE 55W

All four operations will happen, so ask what each one makes possible or safe. Hot hydrogen entering a reactor with oxygen still in it is the reason one step comes first. The oxide has to become metal before anything is active. And a low-flow check happens before production feed is committed to an unknown charge.

ASK 10W

Order the four operations for bringing the charge into service.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

nickel oxide has to be reduced to nickel metal before it is active · hydrogen at temperature will reduce a metal oxide

VERDICT 81W

Each operation makes the next one possible or safer. Nitrogen purge removes oxygen before hot hydrogen enters the reactor. Reduction must then convert nickel oxide into active nickel metal. A low-flow light-off test checks whether the new charge is active before full production feed is committed. Only after that should the plant run a matched comparison against a known good charge. The comparison creates a quantitative baseline for the next crew instead of leaving them only a verbal impression of performance.

TAKEAWAY 18W

A part that cannot be replaced is proved at low rate before it is trusted at full rate.

10.2 One improvement, four candidates

ELECTROLYSIS HALL · A PERSON · SCIENCETANK

SCENE 37W

The station has parts and crew time for one improvement. One catalyst charge is already lost, and there is no spare after the new one. Achebe puts four candidates on the board before anyone starts cutting pipe.

GUIDE 85W

The plant ran 38 sols on water no column was polishing, and lost a catalyst charge to it. Conductivity was logged on every one of those sols and read by nobody. The array has not been the limit on any sol this season. There are already four probes in the catalyst bed. A second column in series lets the lead bed be changed at breakthrough, and needs pipe and a shutdown to fit. One catalyst charge is left, and there is no spare after it.

ASK 12W

Fund the change that would most alter what the plant does next.

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

the polishing column gives no warning as it fills · the plant has no spare catalyst charge after this one · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 86W

The failure was not that the plant lacked a measurement. Conductivity was recorded every sol. The failure was that no rule turned the measurement into an action. A feed-water alarm would have made the breakthrough visible while the catalyst could still be protected. A second polishing column adds capacity but still needs a trigger for changeover. Extra array power does not address the fault, and the catalyst already has several temperature probes. The highest-leverage change closes the gap between an existing signal and an irreversible consequence.

TAKEAWAY 16W

The work worth doing first is the work that would have caught the fault nobody caught.

10.3 The tank pressure that is not a leak

COLD END · A MODULE · CHOICE

SCENE 34W

The methane tank has climbed about 0.3 bar a sol for 4 sols. The mass in the tank is unchanged across the same four sols, and Moreau is being asked whether to vent it.

GUIDE 62W

Four readings of rising pressure at constant mass. Ask of each whether the inventory supports it. A leak or sustained venting takes mass out, and four sols show none gone. A tank holding liquid and vapour sits near the vapour pressure set by its temperature, so pressure is a sensitive thermometer. The response is to find the heat rather than to vent.

ASK 13W

The pressure is rising and the mass is not falling. What is happening?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a liquid in a closed tank sits under the pressure of its own vapour · vapour pressure rises steeply with temperature

VERDICT 83W

A tank holding liquid and vapour together sits near the vapour pressure set by its temperature. Pressure is therefore a sensitive thermometer while both phases remain present. Here the mass has stayed constant for four sols, so a significant leak or sustained venting is not supported by the inventory. The rising pressure instead points to warming. The response is to find the added heat — insulation, a conducting valve or a refrigeration problem — rather than venting propellant merely because the pressure rose.

TAKEAWAY 15W

Pressure in a cryogenic tank is a reading of temperature until the mass says otherwise.

10.4 Worth digging, or worth leaving

WATER PLANT • A MODULE • CALLBACK • BELT

SCENE 42W

The hopper takes one charge at a time and the digging arm has brought in more than it can process. Some of what is out there gives up water when it is heated. The rest costs the same power and returns nothing.

GUIDE 45W

Two bins. What matters is whether the material holds water that heat can drive off — hydrated minerals and buried ice do, and dry basaltic dust does not, however promising it looks. Sort on what the sample gives up, not on where it came from.

ASK 13W

Send each sample to the bin that says whether heating it returns water.

BEHIND THE BUTTON 87W

Why this is the plant's first decision. Every charge costs the same heating energy whatever is in it, and energy is the thing this station has least of. A hopper charged with dry dust has spent a sol's margin on nothing.

What holds water on Mars. Hydrated sulphates and clays hold it chemically, and it comes off at a few hundred degrees. Buried ice holds it as ice. Weathered basalt, glass and windblown fines hold almost none, and they are the majority of what the arm reaches.

TAKES AS READ

some minerals hold water that heating can drive off • intermolecular forces, and why they set a boiling point — taken as read

VERDICT 154W

Water is the input the whole plant runs on, and heating is how it is recovered, so the only question worth asking of a sample is whether heat gets water out of it.

Hydrated sulphates and clays do: the water is held in the mineral structure and comes off at a few hundred degrees. Buried ice does, obviously and at far lower cost. Weathered basalt, volcanic glass and the windblown fines that cover most of the site hold almost nothing, and they look exactly like the material that does — which is why the sorting happens at the yard rather than at the hopper. The cost of getting it wrong is not the sample. It is the charge: the heater draws the same power whatever is in it, energy is the scarcest thing on this station, and a hopper of dry dust has spent a sol's margin to produce steam that was never there.

TAKEAWAY 15W

Every charge costs the same energy, so what is in it is the whole decision.

END OF THE SOL 13W

A part with no replacement is commissioned by procedure rather than by enthusiasm.

The sol everything agreed

PLAN CARD 127W

Sol 301, and for the first time since the spring the plant has run a full 24 hours without anybody being called out. The new charge is holding conversion at 71% , the polishing column is fresh, and the sol's production came in at 14.1 kilograms against a target of 13.8. Wells has asked for the quiet sol to be spent on arithmetic rather than on improvements, which is not a popular instruction. Today you close the energy ledger against what the array actually delivered, settle how full a cryogenic tank may be filled, and follow the carbon through a full pass of the loop. Herrera's view is that a plant nobody understands on a good sol is a plant nobody can fix on a bad one.

BRIEFING 15W

The new charge ran to specification for a full sol and nobody quite believes it.

WHAT YOU DECIDE 11W

Close the plant's own books while there is nothing to firefight.

11.1 The bill for a kilogram of hydrogen

PLANT CONTROL • A MODULE • BALLPARK

SCENE 30W

Demir is closing the sol's energy ledger. The stacks made 6.9 kg of hydrogen. He wants their electrical energy use written beside the array output before he signs the sheet.

GUIDE 58W

Five numbers, and two of them are traps. One is the energy stored in a kilogram of hydrogen, which is not what the stack spent. One is the cell voltage. Ask of each what the bill depends on. Real cells run above the reversible voltage, and the difference becomes heat. Using the stored figure understates the electricity badly.

ASK 11W

Estimate the electrical energy the stacks spent making hydrogen this sol.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

electrical energy can be counted in kilowatt-hours • a stack costs more energy per kilogram than the hydrogen itself carries

VERDICT 82W

A kilogram of hydrogen has two useful energy numbers. One is the chemical energy stored in the hydrogen itself. The other is the electrical energy this stack spends to make that kilogram. The second is larger because real cells operate above the reversible voltage. That difference becomes heat and other losses in the electrolysis system. Here the stack uses about 55 kWh/kg while hydrogen stores about 39.4 kWh/kg. Using the stored-energy number as the production cost would understate the electrical bill substantially.

TAKEAWAY 25W

What a product costs to make and what it stores are two different numbers, and the gap is the price of running at a rate.

11.2 The space left at the top

COLD END · A PERSON · CHOICE

SCENE 32W

Bhatt's loading procedure stops the methane tank at 95% by volume. Moreau has been asked why the remaining 5% is not simply more propellant, with the window as close as it is.

GUIDE 58W

Four reasons to leave a gas space. Ask of each whether it is about the liquid or about the vapour. Heat keeps entering, so vapour keeps forming, and it needs somewhere to go before the relief setting is reached. Fill the tank and a small heat input becomes a venting event before the vehicle has used any fuel.

ASK 11W

Why is the flight tank filled to 95% rather than full?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a liquid at its boiling point turns to vapour when heat reaches it · a closed vessel has a pressure it is rated to

VERDICT 86W

A cryogenic tank always contains some vapour because heat continues to enter. The gas space above the liquid is the ullage. It gives newly formed vapour room before pressure reaches the relief setting. Filling the tank completely with liquid removes that buffer. Heating then causes liquid expansion and vapour formation to drive pressure upward quickly. The relief system must vent material to protect the tank. Leaving planned ullage therefore prevents a small heat input from immediately becoming a propellant-loss event before the vehicle uses the fuel.

TAKEAWAY 14W

A cryogenic tank is designed around what happens to it between filling and use.

11.3 Close the carbon pass

REACTOR HALL · A MODULE · BALANCE

SCENE 35W

Herrera pins one pass of the carbon assay to the board: 100 moles of carbon dioxide entered. Methane and carbon monoxide are measured at the outlet, and the recycle meter records the unreacted return stream.

GUIDE 57W

Read every stream, then count only the carbon that belongs in this pass's ledger. A hundred moles went in; methane and carbon monoxide are measured at the outlet and the recycle meter records what came back unreacted. Whatever the counted streams do not account for is missing, and where it went decides what repeated recycle can achieve.

ASK 21W

Where are all 100 carbon moles after 1 pass, and what does that imply for overall methane yield after repeated recycle?

BEHIND THE BUTTON 143W

Why one pass is not the yield. A single pass through the reactor converts part of the feed. The rest returns and goes round again, so the overall yield after many passes can be far higher than the single-pass conversion — as long as nothing is lost on each circuit.

What counting a stream claims. Reading is free. Counting says this carbon belongs to the same hundred moles that went in. Carbon monoxide and methane are both products; unreacted feed is not a product but is still carbon that must appear somewhere.

Why a gap changes the plan. Carbon that leaves the loop each pass — deposited on the catalyst, vented, or dissolved — is carbon the recycle can never recover. A small per-pass loss compounds into a hard ceiling on overall yield, which is the number the ascent vehicle's schedule depends on.

TAKES AS READ

the recycle loop returns unreacted gas to the inlet · every carbon atom entering the bed leaves it in something

VERDICT 79W

A one-pass conversion of 64% does not mean 36% of the feed is lost. The carbon ledger shows 64 moles leaving as methane, 2 as carbon monoxide and 34 returning as unreacted carbon dioxide. The recycle term is therefore inventory, not waste. If the same selectivity holds over repeated passes, the returned carbon keeps getting another chance. The ultimate split approaches the ratio of the irreversible products: $64/(64+2)$, or about 97% methane, with about 3% ending as carbon monoxide.

TAKEAWAY 18W

Per-pass conversion counts what reacts now; overall conversion counts what is eventually recovered after unreacted feed is recycled.

11.4 Getting water out of frozen ground — Review

WATER PLANT · A MODULE · CALLBACK · SEQUENCE

SCENE 31W

A load has come in at 3% water instead of eight, and the same four operations have to run on it. Nakagawa wants the order confirmed before the hopper is sealed.

GUIDE 55W

All four operations will happen, and this load is lean. Ask of each what would be indistinguishable without it. Without a figure to expect, a poor yield and a plant fault look the same. And the warming has a ceiling as well as a floor, because perchlorate salts break down when they are heated hard.

ASK 7W

Order the operations for a lean load.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

warming ice-bearing soil releases water as vapour at low pressure · a leaner load carries the same salts in less water

VERDICT 82W

Sampling comes first on a lean load for a reason that does not apply to a rich one: without a figure to expect, a poor yield and a plant fault look identical. Warming has a ceiling as well as a floor, since perchlorate salts break down when they are heated hard and release chlorine compounds into the vapour. Measuring what was collected against what was expected is the comparison that makes the sample worth taking, and the polishing rate does not change.

TAKEAWAY 15W

A poorer load changes the yield of a process and not the order of it.

END OF THE SOL 14W

The sol to check the books is the sol when nothing is going wrong.

40% of the sun

BEFORE THE SOL — EVADE

Earth wants a call and the window is closing

There is a downlink slot in twenty minutes and the flight surgeon wants a personal medical conference that will take all of it. The tank pressure survey has to go this window or wait a sol. Keep out of the way until the data is queued.

PLAN CARD 123W

Sol 302. A regional storm crossed the station overnight and the sky this morning is the colour of the ground. Iosif Petrov, the array technician, has the field delivering 430 kilowatt-hours against 2000 on a clear sol, and nobody can say whether that lasts 2 sols or 20. Everything on this plain runs off that array: the stacks, the refrigerators, the drill, and the habitat that Demir keeps at 19 degrees. Today you work out how much light is actually reaching the panels, decide in what order the loads are shed, and settle what the water plant does while the drill is stopped. Wells will sign whatever order you set, and she has said she will not revisit it before the storm clears.

BRIEFING 14W

A regional dust storm arrived overnight and the array is delivering less than half.

WHAT YOU DECIDE 10W

Decide what the station stops doing when the power halves.

12.1 What the dust leaves of the sun

ELECTROLYSIS HALL · A MODULE · BALLPARK

SCENE 33W

Petrov has the optical depth from the station's own sun sensor: 2.4, against 0.4 on a clear sol. Above the dust the beam would deliver 490 watts to each square metre of panel.

GUIDE 60W

Five numbers, and three of them belong to the array rather than to the beam: the efficiency, the area, and the clear-sol depth. Ask of each whether this transmission needs it. Each unit of depth divides the beam again, so the effect is multiplicative rather than subtractive. And note the limit: this is the direct beam, not the array's output.

ASK 10W

Estimate the direct beam reaching a panel through this dust.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

each unit of optical depth cuts a beam by a further factor · a beam is attenuated as an exponential of the depth it passes through · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 83W

Dust attenuation is multiplicative. Each additional unit of optical depth removes the same fraction of the beam that remains. That is why a change from $\tau = 0.4$ to $\tau = 2.4$ is much worse than subtracting 2 equal pieces of sunlight. The direct beam falls to under a tenth of the above-dust value. The array still receives some scattered light from the sky, so direct-beam loss is not the same as total array output. The exponential relation describes only the unscattered beam.

TAKEAWAY 22W

Dust does not dim a beam in proportion; it multiplies it down, so each unit of depth costs more than the last.

12.2 What the station stops doing

PLANT CONTROL · A PERSON · ALLOCATE

SCENE 37W

The dust storm has cut the sol from about 2,000 kilowatt-hours to 430. Tank refrigeration is already drawing power, and six other demands are queued. Demir and Wells want the sacrifices written down before breakers are opened.

GUIDE 60W

Four hundred and thirty kilowatt-hours for the whole sol, against about two thousand on a clear one. Watch the answers list rather than the equipment: it shows what the station can still say at the end of the sol under your plan. Tank refrigeration is already drawing, so the sacrifices have to come from the six demands queued behind it.

ASK 17W

430 kilowatt-hours. Build the sol's plan, and know what it stops the station being able to say.

BEHIND THE BUTTON 134W

Why refrigeration is not negotiable first. Liquid oxygen already made is stored energy the station cannot remake cheaply. Letting it boil off spends propellant that took sols of power to produce, so cutting refrigeration to save power usually loses more power than it saves.

What the answers list is for. Each demand buys the ability to make a particular claim at the end of the sol — that the tank is full, that the assay is current, that the soil work is on schedule. A plan is a choice of which claims survive the storm.

Why writing the sacrifices down matters. Breakers opened without a record become an argument afterwards about what was agreed. Demir and Wells are asking for the list because the storm will last longer than anybody's memory of the reasoning.

TAKES AS READ

stopping a process can be reversible or costly to reverse · a cryogenic tank warms up when its refrigerator stops

VERDICT 86W

The power plan should rank loads by the cost of stopping them. Digging is recoverable because the ground remains available. Batteries can recharge on the next clear sol. Electrolysis can be made up later if schedule margin remains. Cryogenic refrigeration is different. When it stops, heat keeps entering the tanks and already-made propellant boils away. That loss cannot be recovered without making new propellant later. Habitat power is protected separately. The scarce pool should therefore preserve irreversible assets before funding work that can simply be deferred.

TAKEAWAY 16W

Loads are ranked by what stopping them costs to undo, rather than by what they draw.

12.3 Running the plant with no digging

WATER PLANT · A MODULE · CHOICE

SCENE 35W

The drill is off and the hopper is empty. The holding tank still contains 1,100 litres, while the reactor draws about 60 litres per sol. Abiola is asked whether the whole water plant must stop.

GUIDE 61W

Four candidates for what keeps the reactor running. Ask of each whether it stores water or makes it. Eleven hundred litres at sixty a sol is about eighteen sols, which is longer than the storms recorded here. And the reaction itself produces water that the plant condenses and returns. Buffers between steps are what make a plant survive a stopped machine.

ASK 13W

What lets the reactor keep running through a storm with the drill stopped?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the reactor's water demand is met from a holding tank rather than from the drill directly · the Sabatier reaction makes water as well as methane

VERDICT 85W

Two buffers separate the reactor from daily drilling. The first is stored water: 1,100 litres at 60 litres per sol is about 18 sols of inventory. That is longer than the storms recorded at this site. The second is process recycle. The Sabatier reaction produces water, and the plant condenses and returns part of that stream. Those two buffers let extraction pause without forcing an immediate reactor shutdown. They also show why plant reliability depends on inventories between steps, not only on the machines themselves.

TAKEAWAY 17W

A buffer between two steps is what lets one of them stop without the other one noticing.

12.4 Why the hotter pass gave less — Review

REACTOR HALL · A MODULE · CALLBACK · CHOICE

SCENE 33W

Herrera has the equilibrium constant for the methane assembly at four temperatures: 500 K, 550 K, 600 K and 650 K. It falls by more than a factor of 10 across that range.

GUIDE 55W

Four readings of a constant that falls with temperature. Ask of each whether it explains the direction. Falling means heat behaves as one of the products, which is what exothermic means read off a table. Two of these options blame the catalyst or the pressure instead. And the consequence is the plant's whole design problem.

ASK 11W

What does a constant falling steeply with temperature tell the plant?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

the equilibrium constant depends on temperature and on nothing else · this reaction gives out heat

VERDICT 79W

The direction the constant moves in identifies the sign of the heat. Falling with temperature means heat behaves as a product, so heating pushes the balance back — that is the definition of an exothermic reaction read off a table. The practical consequence is the whole design problem of this plant: yield wants a cold bed and rate wants a hot one, and the operating temperature is chosen where a good enough catalyst makes a good enough yield reachable.

TAKEAWAY 14W

A table of constants against temperature is the plant's ceiling written out in advance.

END OF THE SOL 18W

A power cut on a plain with one array is a decision about what the station stops doing.

Three things at once

PLAN CARD 121W

Sol 303, and the storm blew itself out in the last hours of the night. It left three things behind. Moreau's oxygen tank is losing twice the boil-off it lost a fortnight ago; the new catalyst charge has never been checked against the sol-270 run because the storm took the power to do it; and the battery bank came down to 30% overnight and has to be brought back before the next dust event. Today you work out what is wrong with the cold end, decide what a fair comparison of the new charge looks like, and rank all three against each other. Kaur, Moreau and Achebe each want theirs first, and all three of them are being reasonable about it.

BRIEFING 14W

The storm cleared and left three problems that were not there a week ago.

WHAT YOU DECIDE 13W

Rank three faults that arrived together, and act on them in that order.

13.1 Follow the heat to the sky

COLD END · A MODULE · CHAIN

SCENE 35W

Oxygen boil-off has jumped from 12 to 26 kilograms a sol since the storm. Moreau opens the refrigeration schematic beside the tank. Jacket pressure is normal, while the compressor is already at its current limit.

GUIDE 56W

Put the stages in the order the heat actually travels, from the liquid oxygen out to its final sink. Then name the required transfer that is now governing the loss. Jacket pressure is normal and the compressor is already at its limit, so read those as evidence about which link is and is not the problem.

ASK 19W

Trace the heat from the liquid oxygen to its final sink. Which required transfer is now governing the loss?

BEHIND THE BUTTON 131W

Why boil-off is a heat measurement. Liquid oxygen sits at its boiling point, so every joule that reaches it turns some into gas. Twelve kilograms a sol becoming twenty-six means heat is arriving roughly twice as fast, and the tank itself has not changed.

What a compressor at its limit means. The refrigeration chain can only reject as much heat as its weakest stage carries. A compressor already at maximum is not the cause of extra heat arriving; it is the reason none of the extra can be removed.

What the storm changed. Dust reduces the radiator's ability to reject heat to the sky and coats surfaces that were reflecting sunlight. Both push more heat into the same chain, which is why the governing link can move without any component failing.

TAKES AS READ

a refrigerator has to reject its heat somewhere · dust settles on any surface left out in a storm

VERDICT 83W

The refrigerator does not destroy heat. It moves heat from the tank into the refrigerant. Compressor work then raises the refrigerant temperature enough for heat rejection. The condenser transfers that heat to the radiator, which must radiate it to the sky. The vacuum jacket is a separate path for unwanted heat to enter the tank. Its normal pressure makes that path less likely here. Dust weakens the final radiator-to-sky transfer, so the cold end warms even while the compressor works at its limit.

TAKEAWAY 20W

A refrigerator can lose capacity because heat cannot leave its warm end, even when the compressor and insulation are intact.

13.2 Comparing two charges honestly

CATALYST BAY · A MODULE · CHOICE

SCENE 36W

Kaur wants the new charge compared against the sol-270 run before it is trusted with the rest of the season. The plant is at a different temperature, flow and pressure than it was on sol 270.

GUIDE 50W

Four conditions for a fair comparison. Ask of each whether it removes the alternative explanations or adds one. Temperature, pressure and flow are settings; the catalyst is the thing being compared. Running each at its own best condition answers a different question and gives two numbers that cannot be subtracted.

ASK 14W

What has to be true for a comparison between two charges to mean anything?

BEHIND THE BUTTON 195W

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TAKES AS READ

conversion depends on temperature, pressure, flow and the catalyst together · a comparison is only about the variable that was deliberately changed · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 79W

Conversion depends on several variables at once. Temperature, pressure and flow are operating settings, while catalyst condition is the property being compared. If the settings differ between two runs, the comparison mixes catalyst performance with operating effects. Matching those settings removes the main alternative explanations. The remaining conversion difference can then be attributed more credibly to the charge. Running each catalyst at its own best condition answers a different question and produces two results that cannot be cleanly subtracted.

TAKEAWAY 10W

A comparison is worth what its matched conditions are worth.

13.3 Which of the three cannot wait

PLANT CONTROL • A MODULE • CHOICE

SCENE 32W

The three problems are on the board with the power to fix one of them today. Wells wants the order and the grounds for it in the log before anybody starts work.

GUIDE 53W

Three jobs and four options. Ask of each whether the delay costs time or costs a thing already made. Batteries recharge on the next clear sol. Information waits at no cost while the bed is watched. Fourteen extra kilograms a sol is propellant that has already been produced and does not come back.

ASK 8W

Which of the three is dealt with first?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

propellant vented from a tank cannot be recovered • a comparison run and a battery charge can both be done later

VERDICT 80W

Rank by what a delay costs and whether it can be recovered. Boil-off is a loss of a thing already made, running at 14 extra kilograms a sol, and no later work brings those kilograms back. Batteries recharge on the next clear sol. A comparison run is information, and information waits at no cost provided the bed is watched. Only one of the three is spending an asset while it is queued, and that is the one that goes first.

TAKEAWAY 9W

What cannot be undone goes first, whatever is loudest.

13.4 What the plant is protecting now

PLANT CONTROL · A PERSON · SPOT

SCENE 39W

Loads come up on the board through the sol. The station runs to one power rule, and Demir changes it as the sol does — after the dust thickens, after the tank pressure rises, after the window gets closer.

GUIDE 59W

Keep the loads the current rule protects and let the rest drop. The rule at the top of the board is changed during the sol and nobody announces it. What is scored is the loads either side of a change, because they are the only ones that show whether the desk is reading the rule or running yesterday's plan.

ASK 11W

Protect what the current rule protects, and keep watching the rule.

BEHIND THE BUTTON 99W

Why the rule changes at all. On a clear sol the plant is protecting production, because the tank has a schedule. In a storm it is protecting the crew and the propellant already made, because neither can be remade. Close to the window it is protecting anything the launch depends on, which is a third list.

Why the changeover costs. A desk running the production rule through the first hour of a storm keeps the stacks alive and lets the liquefier go, and the propellant that boils off is propellant that has to be made again from the beginning.

TAKES AS READ

a station with limited power keeps some loads and drops others

VERDICT 133W

Three rules run across the sol and each protects a different part of the board. Production, then anything that cannot be remade, then anything the launch window depends on. A load can answer two at once, which is what makes the switch cost real rather than notional, and what the panel scores is the window either side of each change. Everything else on the board is protected by neither rule and is correctly dropped by somebody who has read nothing. The station's own version is the first hour of a dust storm: the rule has changed to protecting what cannot be remade, and a desk still running the production rule keeps the stacks warm while the liquefier goes off — which boils away propellant that took forty sols of array output to make.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

13.5 How much gas the bed has to be fed

REACTOR HALL · A MODULE · CALLBACK · BALLPARK

SCENE 31W

The plant has to average 13.8 kilograms of methane a sol from here to the window. Herrera wants the carbon dioxide that implies before the intake compressors are asked for anything.

GUIDE 59W

Five numbers, and two of them belong to other parts of the equation: the hydrogen ratio, and the water produced. Ask of each whether this carbon bookkeeping needs it. One carbon atom goes from each carbon dioxide into one methane. So the mole ratio is one to one and the mass ratio is not, because the molar masses differ.

ASK 11W

Estimate the carbon dioxide the reactor must be fed each sol.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

one mole of carbon dioxide yields one mole of methane in this reaction · a molar mass converts moles to kilograms

VERDICT 76W

Stoichiometry is mole bookkeeping. The balanced reaction sends one carbon atom from each carbon-dioxide molecule into one methane molecule. Mass adds one conversion step. Carbon dioxide has a molar mass of 44 g/mol, while methane is 16 g/mol. Each kilogram of methane therefore needs 44/16 kilograms of carbon dioxide. The intake compressors must gather that feed from a very thin atmosphere. The required compressor work scales directly with the production target the plant promises to meet.

TAKEAWAY 20W

Stoichiometry turns a production target into a feed rate, and the feed rate is what the intake has to meet.

END OF THE SOL 15W

Three reasonable people wanting to be first is a ranking problem rather than an argument.

Loading is one way

PLAN CARD 122W

Sol 304, and the batch made since the new charge went in is standing in the transfer tank this morning with its assay beside it. It is 410 kilograms of methane, which is 3 sols of the season, and 1 line of the assay sits outside what Bhatt's specification allows. Once it goes across into the vehicle it cannot be taken out again, cleaned, or tested by firing. Today you read the batch against the specification, work out what mass of oxygen the vehicle needs beside it, and decide whether the stacks can be pushed to close the remaining gap. Bhatt has said she will sign what the assay supports and nothing else, and Sundqvist has counted how many sols are left.

BRIEFING 17W

The batch is made, the assay is back, and one line of it is outside the specification.

WHAT YOU DECIDE 8W

Decide whether this batch goes into the vehicle.

14.1 The line that fails

REACTOR HALL • A MODULE • PROTOCOL

SCENE 35W

The loading crew is waiting at the methane umbilical. Bhatt has four assay lines beside four flight limits. Three are inside specification; one is not. Herrera wants the consequence named before the transfer valve opens.

GUIDE 57W

Four assay lines and four consequences. Pair them by asking where each substance would end up as the stream gets colder, and whether that place has moving parts. A specification is not one purity score. Only one of these is outside its own limit, and where it freezes is what makes it the failure rather than ballast.

ASK 11W

Match each assay line to what it means for the vehicle.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

water freezes at a temperature the vehicle's valves reach in use • carbon monoxide does not burn to give the energy methane does • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 83W

A specification is not one overall purity score. Each line protects a different downstream function. Water is the failure here: 400 ppm is 8 times a 50 ppm limit. At cryogenic valve temperatures, that water can become solid and interfere with moving hardware. Carbon monoxide and carbon dioxide are still inside their stated limits. If either rose too far, it would create a different consequence. The vehicle therefore needs every assay line interpreted against its own limit and failure mode before loading begins.

TAKEAWAY 18W

A specification is a list of chemical limits tied to different downstream consequences, not a single purity score.

14.2 How much oxygen a tonne of methane needs

COLD END · A PERSON · BALLPARK

SCENE 29W

Bhatt wants the oxygen mass checked independently before the loading order is written. Methane burns with oxygen 2 moles to 1, and the vehicle carries 6.6 tonnes of fuel.

GUIDE 58W

Five numbers, and two of them belong elsewhere: the methane the vehicle carries, and the molar mass of carbon dioxide. Ask of each whether this ratio needs it. A balanced equation gives moles and tanks are filled by mass, so the molar masses do the converting. And note that an engine may deliberately run away from this ratio.

ASK 12W

Estimate the mass of oxygen needed for each kilogram of methane burned.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a balanced equation gives the mole ratio for a reaction · a molar mass converts a number of moles into a mass

VERDICT 83W

A balanced equation gives a mole ratio, while tanks are filled by mass. Complete methane combustion uses two moles of oxygen for every mole of methane. 2 moles of O_2 have a mass of 64 g, while 1 mole of CH_4 has a mass of 16 g. The stoichiometric oxygen-to-fuel mass ratio is therefore 4:1. A rocket engine may deliberately operate away from that exact ratio for performance reasons. That design choice does not change the stoichiometric calculation the loading team must understand.

TAKEAWAY 15W

A mole ratio is not a mass ratio until both molar masses have been applied.

14.3 What limits the current

ELECTROLYSIS HALL · A MODULE · CHOICE

SCENE 28W

Sundqvist wants the stacks run 20% above their rated current for the sols that remain. Achebe has the cooling loop's capacity written on the board beside the request.

GUIDE 58W

Four candidates for the ceiling on current. Ask of each whether it actually rises with current. Faraday's law puts no ceiling on it at all: more charge is simply more product. What rises is the wasted voltage. Resistance and slow electrode steps both cost more when the cell is driven harder, and that becomes heat inside the stack.

ASK 11W

What sets the highest current these stacks can be run at?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the product a cell makes is proportional to the charge it passes · the voltage above the minimum leaves the cell as heat

VERDICT 78W

Faraday's law puts no ceiling on current at all: more charge is simply more product, in exact proportion. What rises with current is the overpotential, because resistance and sluggish electrode steps both cost more voltage the harder the cell is driven, and that extra voltage becomes heat inside the stack. So the limit is set by what the cooling loop can carry away before the cell cooks its own seals. 20% more current can cost 40% more heat.

TAKEAWAY 11W

The ceiling on a cell is usually thermal rather than electrochemical.

14.4 What comes with the water

WATER PLANT • A MODULE • CALLBACK • PROTOCOL

SCENE 35W

The holding tank's analysis is on Abiola's bench: four dissolved species, none of them water. Each one has somewhere downstream that it matters, and the plant has one polishing column between them and everything else.

GUIDE 48W

Four dissolved species and four consequences. Pair them by asking what surface or process each one reaches downstream. One binds to nickel. One deposits on hot faces. One fouls mechanically. And one is already handled as a process gas, which is why not every impurity is equally dangerous.

ASK 9W

Match each dissolved species to what it does downstream.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

salts dissolve into ions in water • an electrolysis cell and a catalyst bed both have surfaces that can be fouled • intermolecular forces, and why they set a boiling point — taken as read

VERDICT 76W

A dissolved species matters because of the surface or process it reaches downstream. Chloride is dangerous here because it can bind strongly to nickel catalyst sites. Sulphate can contribute to scale on hot surfaces. Fine particles create a mechanical fouling problem. Dissolved carbon dioxide is different because the plant already handles carbon dioxide as a process species. The polishing train should therefore be designed around consequences, not around the idea that every impurity is equally harmful.

TAKEAWAY 14W

Water from the ground is a solution, and every solute has an address downstream.

END OF THE SOL 15W

A decision that cannot be reversed is worth more care than one that costs more.

What the next crew is told

BEFORE THE SOL — TAG

Catch Achebe before the array goes into storm mode

Achebe, the power and electrolysis engineer, has the only measurement of what the array actually delivered before the dust thickened, and once it folds into storm mode the sensors read nothing useful for days. Every power decision this week is against that number.

PLAN CARD 120W

Sol 305, the last sol of this rotation, and the handover file has to be finished this morning. The plant is making 14.4 kilograms a sol against a target of 13.8, the tank stands at 4.3 tonnes of methane, and the arithmetic now finishes the fill 11 sols before the window opens. None of that is the difficult part. Today you decide what is left funded and instrumented for the crew who arrive on the next vehicle, write the standing order for the water loop, and say honestly what is known about the catalyst and what is not. Wells has one instruction about the file: that somebody who was not here can act on it without asking anybody who was.

BRIEFING 12W

The fortnight ends, and what survives it is what gets written down.

WHAT YOU DECIDE 14W

Say what is known, how well it is known, and what was never established.

15.1 What to leave running

PLANT CONTROL · A MODULE · SCIENCETANK

SCENE 36W

Only 2 improvements can be left in place for the crew arriving 300 sols from now. Wells and Demir have four candidates on the final board. The failed catalyst charge is on the floor beside them.

GUIDE 76W

This season lost 38 sols of production to a measurement taken every sol, logged every sol, and read by nobody. The bed already carries four thermocouples. The array has one sun sensor, and it has never disagreed with the sky. The next crew arrives 300 sols from now and will not have met anybody who was here. An alarm reaches somebody without being looked for. A written order tells whoever hears it what to do next.

ASK 7W

Fund what the next crew should inherit.

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

the fault that cost this season went undetected because nothing measured it · the next crew arrives without anybody from this one

VERDICT 85W

The best inheritance is a system that turns a known failure mode into an automatic response. The conductivity alarm closes the exact gap that let breakthrough continue unnoticed. The standing order for lead-and-lag column changeover tells the next crew what to do when that signal arrives. Spare thermocouples are useful but the bed already has temperature coverage, and a second sun sensor does not address the failure that consumed the catalyst. Together, alarm plus procedure connect measurement, threshold and action without requiring this crew's memory.

TAKEAWAY 12W

What outlives a crew is what somebody made automatic before they left.

15.2 Write the rule before the trace moves

WATER PLANT · A PERSON · TRIGGER

SCENE 35W

Abiola is writing a rule for a crew he will never meet. He loads last season's conductivity trace into the alarm simulator. Once the replay starts, the thresholds are locked and the values run forward.

GUIDE 72W

One rule, written before the trace moves and fixed once you release it. The changeover takes one update from the moment it is called, and past $3 \times$ baseline the feed water is outside the accepted range. So the line goes back from 3 by however far the trace climbs in that update — and not so far back that a clean bed is swapped out for a reading which meant nothing.

ASK 20W

At what conductivity do you move the lag column into lead service, given the changeover takes one update to complete?

BEHIND THE BUTTON 132W

Why conductivity is the signal. It rises as the column loses capacity and lets ions through, so it is an early indicator of exhaustion rather than a measurement of failure. The trace moves before the water is out of specification, which is what makes a rule possible at all.

Why two thresholds. A warning gives somebody time to prepare the fresh column; the changeover commits the swap. One threshold doing both means the preparation starts at the moment the swap should already have happened.

What writing for an absent crew changes. No judgement is available at the moment it fires, so the rule has to hold for a trace nobody has seen, including a fast rise. That is why the margin is set against the slowest case rather than the typical one.

TAKES AS READ

an exchange bed reaches breakthrough without warning · a lead and lag pair lets one bed be changed while the other works

VERDICT 85W

The outlet can remain near baseline while the exchange front is still inside the bed, then rise quickly as the front reaches the end. A threshold is useful only if the response can finish before the water leaves specification. Here changeover takes one update. Triggering at twice baseline completes the swap while the next point is still 2.6 times baseline. Waiting for 2.6 completes it after the trace has crossed the 3.0 limit. Precommitting the rule prevents the next crew from turning hindsight into policy.

TAKEAWAY 18W

A breakthrough threshold must be written before the rise and early enough to cover the action's lead time.

15.3 The honest claim about the bed

CATALYST BAY · A MODULE · CHOICE

SCENE 35W

Kaur is writing the final catalyst handover. The bed has run for five sols at one set of conditions using water from a fresh polishing column. It is the last charge available on the station.

GUIDE 49W

Four claims about one charge. Ask of each whether the five sols support it. One set point on clean water says something narrow about that window. It says nothing about other temperatures or flows. And it cannot test chloride resistance, because chloride was deliberately absent after the column change.

ASK 11W

What is the honest claim to hand over about this charge?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a claim is limited by the conditions it was tested under · a poison accumulates over time rather than acting at once · rate constants, and how steeply they climb with temperature — taken as read

VERDICT 88W

A claim is limited by the conditions under which the evidence was collected. Five sols at one set point on clean water support a narrow statement about that operating window. They do not establish performance at other temperatures or flows. They also do not test resistance to chloride because chloride was intentionally absent after the column change. A useful handover keeps those conditions attached to the claim. The next crew can then tell whether a future observation is a replication of this result or a genuinely new test.

TAKEAWAY 16W

The strength of a claim is set by what was varied while it was being tested.

15.4 Where 8% of the charge went

ELECTROLYSIS HALL · A MODULE · CALLBACK · BALANCE

SCENE 35W

The cell held 640 amps for a full sol. Faraday's law predicts about 594 grams of hydrogen, but the separator collected only 546. Achebe has already checked the collection-line pressure and the hydrogen inventory downstream.

GUIDE 56W

Read every stream, then count only what belongs in the charge ledger. Faraday's law says 640 amps for a sol should give about 594 grams of hydrogen and the separator collected 546. Achebe has already cleared the collection line and the downstream inventory, so the missing 8% is a term the ledger does not yet have.

ASK 15W

What term closes the charge ledger, and what does that term measure about the cell?

BEHIND THE BUTTON 131W

What Faraday's law is claiming. Every electron that crosses the cell makes a fixed amount of hydrogen. So the predicted mass is arithmetic on the charge, and it is an upper bound: it assumes every electron did the intended reaction.

Why the gap is a measurement rather than an error. Electrons that did something else — a side reaction, a shuttle current, recombination inside the cell — still crossed and still counted as charge. The shortfall is therefore a measurement of how much of the current did useful work, which is the cell's current efficiency.

Why checking the line first was right. A leak or unmeasured inventory downstream would explain the gap without saying anything about the cell. Ruling those out is what turns a discrepancy into evidence about the electrochemistry.

TAKES AS READ

Faraday's law converts charge into a theoretical amount of product · current efficiency is collected product divided by theoretical product from charge

VERDICT 79W

Current times time gives charge, and Faraday's law converts that charge into a theoretical hydrogen mass. For this run the charge corresponds to about 594 grams. Only 546 grams were collected, while line inventory and measured vent loss were unchanged. The missing 48 grams are therefore not a hidden hydrogen store. They are hydrogen-equivalent charge that crossed by other electrochemical paths or recombined before collection. The ratio 546/594 is about 92%, which is the current efficiency for this run.

TAKEAWAY 20W

Faraday's law fixes the theoretical product from charge; current efficiency measures how much of that charge reached the intended reaction.

A handover states what is known, how well, and what was never established at all.

THE CLOSING CARD

The fill finished on sol 475, eleven sols before the window opened, with six point six tonnes of methane and twenty-three of oxygen aboard and every batch signed against the assay it came with. What had been holding the plant down was never the set point both sides spent a fortnight arguing about: the polishing column filled around sol 261 and passed chloride to the reactor for thirty-eight sols, and the catalyst died from the inlet end while the temperature profiles were being read as heat damage. The last spare charge went in on sol 300 and held 71% conversion to the end.

What it cost: eleven sols of production, one catalyst charge with no replacement on the planet, and about four hundred kilograms of oxygen out of the vent while a radiator stood under a fortnight of storm dust. What is unfinished: the lead-and-lag columns are plumbed and the interval between changes is a guess taken from one breakthrough; the conductivity alarm was wired to a panel four sols before the rotation ended and has never sounded; and nobody has measured what this ground does to a catalyst at the deeper cut, which is the ice the next crew will be working.

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